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Probing Growth-Induced Anisotropic Thermal Transport in High-Quality CVD Diamond Membranes by Multifrequency and Multiple-Spot-Size Time-Domain Thermoreflectance

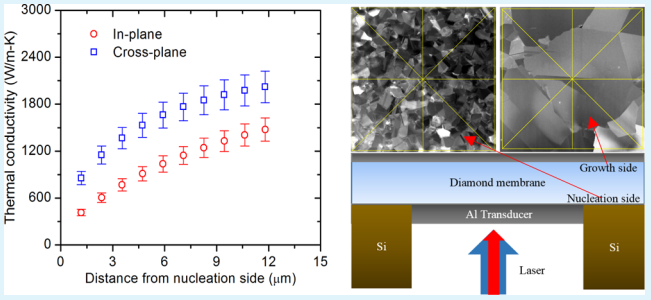
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ABSTRACT: The maximum output power of GaN-based high-electron mobility transistors is limited by high channel temperature induced by localized self-heating, which degrades device performance and reliability. Chemical vapor deposition (CVD) diamond is an attractive candidate to aid in the extraction of this heat and in minimizing the peak operating temperatures of high-power electronics. Owing to its inhomogeneous structure, the thermal conductivity of CVD diamond varies along the growth direction and can differ between the in-plane and out-of-plane directions, resulting in a complex three-dimensional (3D) distribution. Depending on the thickness of the diamond and size of the electronic device, this 3D distribution may impact the effectiveness of CVD diamond in device thermal management. In this work, time-domain thermoreflectance is used to measure the anisotropic thermal conductivity of an 11.8 μm-thick high-quality CVD diamond membrane from its nucleation side. Starting with a spot-size diameter larger than the thickness of the membrane, measurements are made at various modulation frequencies from 1.2 to 11.6 MHz to tune the heat penetration depth and sample the variation in thermal conductivity. We then analyze the data by creating a model with the membrane divided into ten sublayers and assume isotropic thermal conductivity in each sublayer. From this, we observe a two-dimensional gradient of the depth-dependent thermal conductivity for this membrane. The local thermal conductivity goes beyond 1000 W/(m K) when the distance from the nucleation interface only reaches 3 μm. Additionally, by measuring the same region with a smaller spot size at multiple frequencies, the in-plane and cross-plane thermal conductivities are extracted. Through this use of multiple spot sizes and modulation frequencies, the 3D anisotropic thermal conductivity of CVD diamond membrane is experimentally obtained by fitting the experimental data to a thermal model. This work provides an improved understanding of thermal conductivity inhomogeneity in high-quality CVD polycrystalline diamond that is important for applications in the thermal management of high-power electronics.



KEYWORDS: *CVD diamond, grain boundary, anisotropic thermal conduction, nonhomogeneous conduction, electronics cooling, time-domain thermoreflectance*

1. INTRODUCTION

The development of GaN-based high-electron mobility transistors (HEMTs) has much promise for creating advanced communications and radar technology.[1] However, localized power densities in GaN HEMTs can exceed 90 kW/cm^2, resulting in high channel temperatures that degrade device performance and reliability.[2,3] The integration of high thermal conductivity chemical vapor deposition (CVD) diamond into GaN HEMTs allows for effective heat spreading and the ability of keeping these devices cool.[4−8] An innovative way of integrating the CVD diamond with GaN HEMTs is through the growth of the diamond onto the backside of the GaN buffer layer. When growing CVD diamond, substrates are mechanically abraded and seeded with diamond powder. The diamond then grows from these seeds, in a columnar grain structure, expanding laterally as the film thickness increases from the growth interface. The columnar grain structure makes heat transport in the cross-plane direction different from that in the in-plane direction. The thermal conductivity also varies along the growth direction because both in-plane and cross-plane crystal sizes increase with increasing distance from the nucleation sites. The crystal size on the growth side is much larger than that on the nucleation side. Consequently, polycrystalline diamond membranes show three-dimensional

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(3D) anisotropy of thermal conduction because of their inhomogeneous structure.[9−13]

The anisotropic heat conduction in CVD diamond has attracted great attention because it affects the ability of extracting heat through conduction.[9,13−24] However, almost all experimental measurements have been focused on two-dimensional (2D) anisotropy of thermal conduction.[9,13−21] Three-dimensional anisotropy has not been reported before experimentally due to difficulties in thermal measurements. TDTR is a popular noncontact optical pump and probe thermal characterization method used to measure thermal properties of both bulk and nanostructured materials.[25−29] A modulated pump beam heats a sample periodically, and a probe beam measures the temperature of the sample surface through a change in reflectance (i.e., thermoreflectance). The probe beam delay time is controlled by a mechanical stage, which is used to create a temperature decay curve from 0.1 to 5 ns. By fitting the experimental signal picked up by a lock-in amplifier with a multilayer thermal model, thermal properties of the sample can be extracted. In time-domain thermoreflectance (TDTR) measurements, the distance heat penetrating into the surface depends on the modulation frequency and the thermal diffusivity of the sample. By tuning the modulation frequency, we can infer the thermal properties of the sample with different penetration depths, which provides an excellent nondestructive way to explore the 3D anisotropy of inhomogeneous CVD diamond membranes.

In this work, we measured thermal properties of a CVD diamond membrane from the nucleation side by multi-frequency and multiple-spot-size TDTR. Following deposition of the diamond film, the supporting substrate was etched selectively to fabricate suspended diamond membranes.[12] The membranes were coated with an Al layer as a transducer for TDTR measurements. Measurements with modulation frequencies from 1.2 to 11.6 MHz were performed with a pump radius of 19.8 μm to measure the thermal properties in the cross-plane direction. On the same spot, measurement with modulation frequencies from 3.6 to 6.3 MHz was performed with a pump radius of 5.3 μm to extract information on the in-plane and cross-plane thermal conductivities. After measuring the grain size distribution of nucleation and growth sides with a transmission electron microscope (TEM), a thermal conductivity model was used to understand the experimental results and obtain the 3D anisotropic thermal conductivity. This work provides insight toward an improved understanding of heat conduction inhomogeneity in CVD polycrystalline diamond membranes for applications of heat dissipation in high-power electronics.

2. SAMPLE DESCRIPTION AND CHARACTERIZATION

CVD diamond membranes were grown on a silicon wafer by Element Six Technologies with a diamond thickness of 11.8 μm. By using a mixture of hydrogen and methane, diamond was grown on intrinsic ⟨100⟩ silicon substrates by microwave-enhanced CVD technique. To facilitate diamond growth, silicon wafers were mechanically abraded and seeded with diamond powder. Further information can be found in ref 30. The definition of the growth and nucleation sides is provided in Figure 1a. The nucleation side is adjacent to the silicon substrate, and the growth side is in contact with the vapor during the growth process. The silicon substrate was etched away with plasma, and both the growth and nucleation sides of the diamond were coated with an Al thin film to serve as the

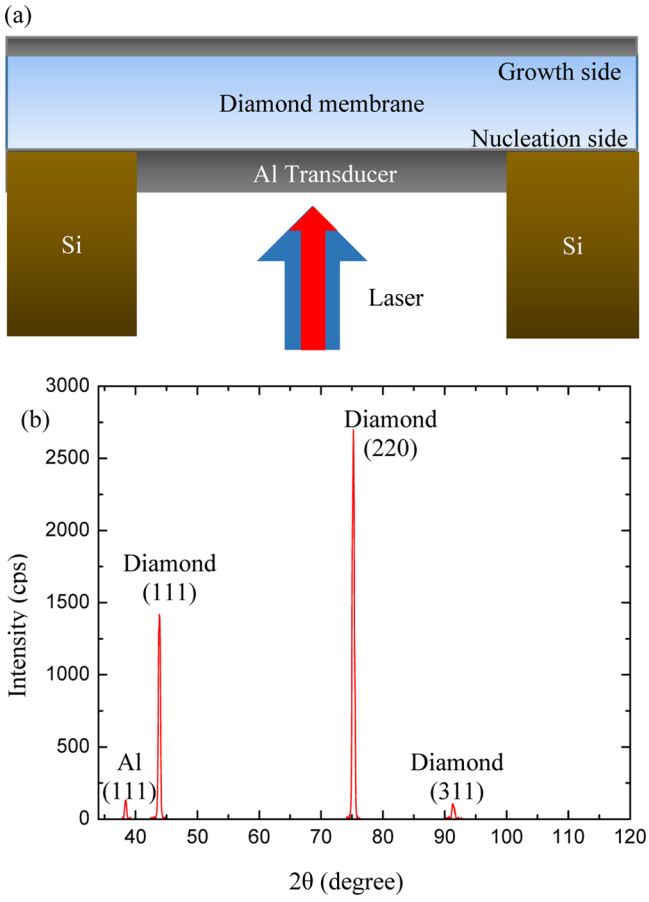


Figure 1. (a) Schematic diagram of the sample structure. The silicon substrate is etched and Al transducer is coated on both the growth and nucleation sides. Pump (blue) beam heats the sample, and probe (red) beam measures the temperature variation. (b) XRD pattern for the diamond membrane coated with a layer of Al. Both Al and diamond peaks show up in the pattern.

transducer layer for subsequent TDTR measurements. The Al thicknesses were determined by a picosecond acoustic method in the TDTR experiments as 103 and 218 nm on the growth and nucleation sides, respectively.

The Al layer on the growth side was deposited by E-beam evaporation. Owing to the rough surface of the growth side, we did not perform TDTR measurements on the growth side. All TDTR measurements in this work were performed on the nucleation side. We deposited a thick Al layer (218 nm) on the nucleation side by sputtering deposition and obtained high sensitivity for Al thermal conductivity. The Al thermal conductivity was measured as 152 W/(m K) by multifrequency TDTR measurements and fixed in the subsequent anisotropy thermal model. We calculated the steady-state temperature rises in these measurements and found all of them to be very small (less than 1 K). The sensitivity of the Al−diamond interface thermal conductance on the growth side is very small, so we fix it as 100 MW/(m^2 K) in our data fitting.[31,32] The pump and probe beam radii measured by a DataRay scanning slit beam profiler are 19.8 and 7.3 μm, respectively, when using the 5× objective and 5.3 and 2.1 μm, respectively, for the 20× objective.

X-ray diffraction 2θ:ω scans were performed on a Bruker JV D1 diffractometer with Cu Kα_1 radiation and a parallel beam source. The acceptance angle for the diffracted beam was

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~0.36°. The sample was mounted vertically on a background-free stage. In the $2\theta{:}\omega$ scan, ω was offset by 10° from the surface orientation of Si substrate to avoid Si. Figure 1b shows the XRD pattern of the diamond membrane coated with a layer of Al. Peaks of both Al and diamond show up in the pattern. Additionally, a focused ion beam (Nova 600 FIB) was used to prepare plan-view samples before they were characterized by a Titan 300 S/TEM (FEI) to obtain the grain morphology of the nucleation and growth sides of the diamond membrane. The scanning transmission electron microscopy (STEM) mode in TEM with high angle annular dark field detector reveals contrast from different grains. Figure 2 shows the plan-view TEM images of the grain structure on the nucleation and growth sides. The dimensions of yellow squares in Figure 2a,b are 1.57 μm × 1.57 μm and 6.27 μm × 6.27 μm, respectively. As shown in Figure 2a, a yellow square with a side length of 1.57 μm is placed on the TEM plan-view picture. A cross-line is used to connect the middle points of the four sides, and another cross-line is used to connect the corners of the square. After measuring the crystal sizes along these four lines, we obtain the average grain size. The grain sizes of the nucleation and growth sides of the diamond membrane were quantified (ASTM E-112, Standard test methods for determining average grain size, 2010). A summary of the grain size distributions is shown in Figure 3. The grain size of the growth side is much

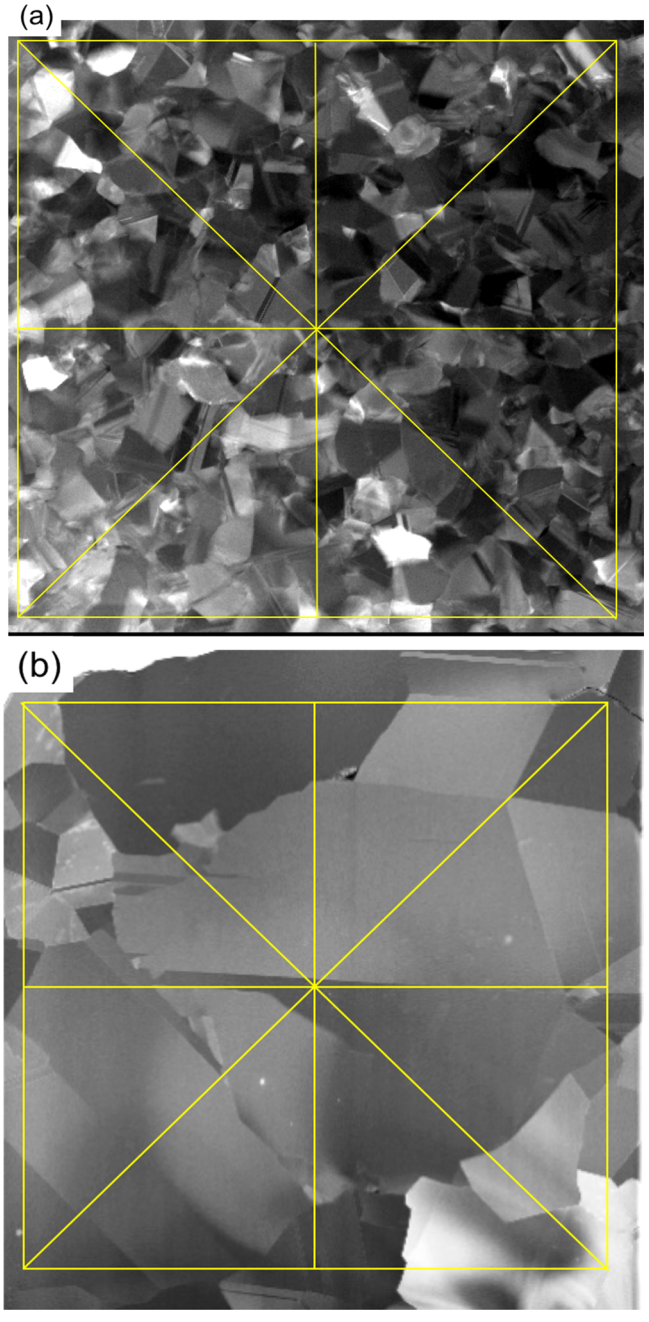


Figure 2. Plan-view grains of nucleation (a) and growth (b) sides of diamond membrane measured by TEM. The dimensions of yellow squares in (a) and (b) are 1.57 μm × 1.57 μm and 6.27 μm × 6.27 μm, respectively.

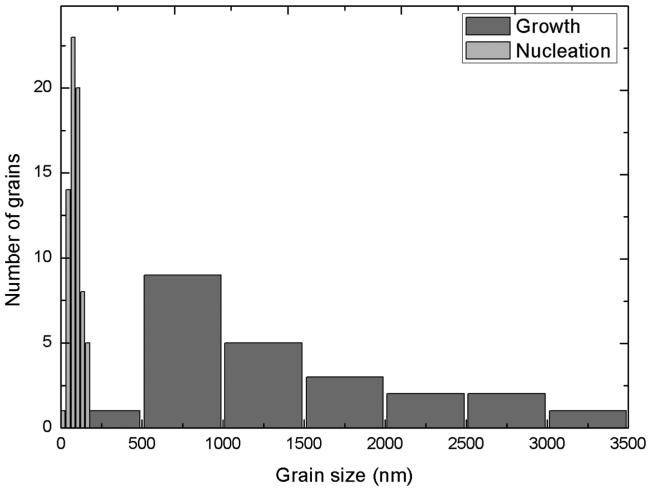


Figure 3. Grain size distributions of nucleation and growth sides of the diamond membrane.

larger than that of the nucleation side. For the nucleation side, most of the grain sizes are in the range of 60−120 nm with an average grain size of 92 nm. For the growth side, most of the grain sizes are in the range of 500−1500 nm with an average value of 1360 nm. TDTR was measured from the nucleation side of the diamond membrane. The pump and probe beams cover 13275 and 2084 grains even for the 20× objective. The beam size is much larger than the grain size. Therefore, we think that the effect of spot variation on TDTR measurements is negligible in this case.

3. RESULTS AND DISCUSSION

3.1. Gradient Thermal Conductivity Model. To understand the inhomogeneous structure and thermal properties of the CVD diamond membrane, a gradient thermal model, similar to that presented by Sood et al., is applied.[9] A schematic diagram for the nature of crystal growth in CVD diamond membranes is shown in Figure 4. Starting at the seed layer, grains are assumed to grow laterally in size as the thickness of the film increases. As they grow, adjacent grains will contact and compete to continue their growth process. The thickness of the diamond membrane is 11.8 μm, which for the model is divided into 10 sublayers, each with a different isotropic thermal conductivity (κ_1−κ_{10}). In CVD diamond membranes, the in-plane grain size increases approximately linearly with the distance from the nucleation interface when the diamond thickness is less than 100 μm.[9,33] The in-plane grain size follows a linear relationship with film thickness

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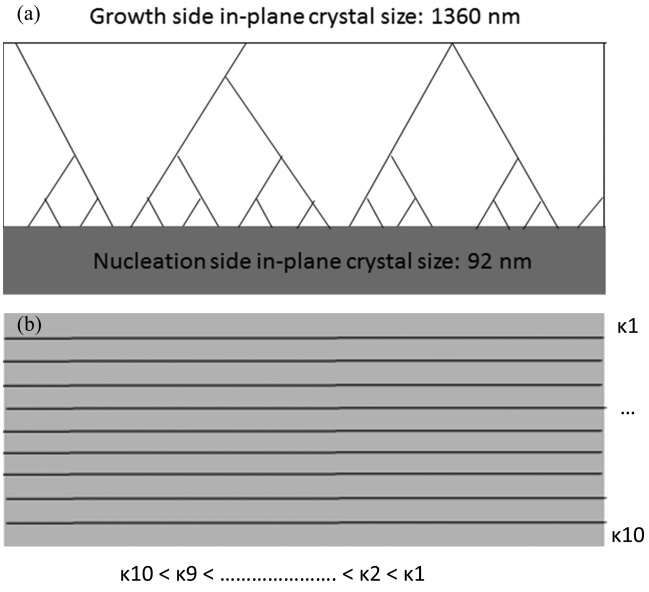


Figure 4. (a) Schematic diagram of crystal growth in CVD diamond. Total thickness of the diamond membrane is 11.8 μm. (b) Membrane is divided into 10 sublayers with different isotropic thermal conductivities $\kappa 1$, $\kappa 2$, ..., $\kappa 10$.

$$L_{in}(z) = d_0 + \alpha \times z$$

where d_0 is the in-plane crystal size in the nucleation side, α is a constant, and z is the distance from the nucleation interface. Cross-plane grain sizes are much larger than in-plane grain sizes. Here, for simplicity, we assume that the diamond is isotropic in each single sublayer and, to the first-order approximation, define an effective grain size as $L_{eff}(z) = A \times L_{in}(z)$. Here, the constant A includes the effect of anisotropy, phonon transmission, and specularity. We neglect phonon-defect scattering and only consider grain boundary scattering and phonon−phonon scattering as the dominant scattering sources, which limit the phonon mean free path. Thus, the phonon mean free path, after considering size effects according to Matthiessen's rule, is

$$\lambda(z) = \left(\frac{1}{L_{eff}(z)} + \frac{1}{\lambda_{bulk}}\right)^{-1} \quad (1)$$

Here, λ_{bulk} is the phonon mean free path in bulk diamond. In diamond, 80% of the heat is carried by phonons with mean free path from 550 to 3400 nm.[34] Similar to ref 9, we take 1 μm as the frequency-independent bulk phonon mean free path (gray approximation), where the phonon free path accumulation function shows the steepest increase. Then, the thermal conductivity after considering the size effect is

$$\kappa_{size}(z) = \frac{\lambda(z) \times \kappa_{bulk}}{\lambda_{bulk}} \quad (2)$$

Here, κ_{bulk} is the thermal conductivity of single crystal diamond. We take the theoretical value of 3000 W/(m K) based on the first-principles calculation.[9,34] The thermal interface resistance between grains also needs to be considered. The corresponding thermal conductivity is

$$\kappa(z) = \left(\frac{1}{\kappa_{size}(z)} + \frac{R_{gb}}{L_{eff}(z)}\right)^{-1} \quad (3)$$

Here, thermal interface resistance R_{gb} takes the value of 0.1 m^2 K/GW according to simulations in refs 35 and 36 and our data fitting, which will be discussed later.

Multifrequency measurements on the same spot on the nucleation side were conducted from 1.2 to 11.6 MHz with 5× objective. If isotropy is assumed when fitting the data, we can obtain frequency-dependent effective thermal conductivity, as shown in Figure 5. The effective thermal conductivity decreases

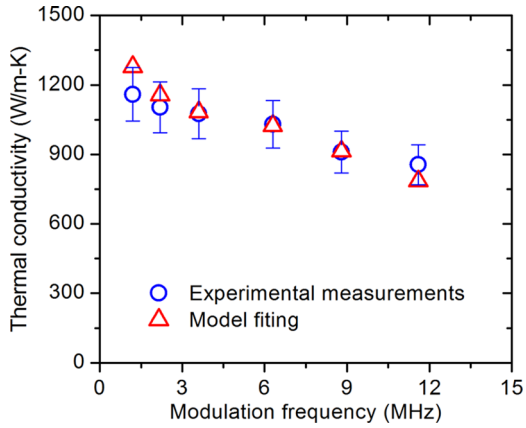


Figure 5. Frequency dependence of thermal conductivity of the diamond membrane.

with increasing frequency. At low modulation frequencies, the penetration depth of the thermal wave is quite large and results in the sampling of more high thermal conductivity diamond further from the nucleation side of the membrane, thereby yielding a larger effective thermal conductivity.

For the gradient thermal model discussed previously, constants A and R_{gb} are unknown parameters, which can be determined by fitting the model to the frequency-dependent effective thermal conductivity. An iterative fitting procedure is employed, where an initial guess is made for the two values of A and R_{gb}. Using this initial guess for A and R_{gb}, the 10-layer gradient thermal conductivity is calculated and used to generate a theoretical TDTR−V_{in}/V_{out} curve. We then fit this theoretical curve with an isotropic three-layer (Al−diamond−Al) TDTR model to obtain an effective thermal conductivity for the sampled volume of the membrane. After comparing this calculated value to the measured effective thermal conductivity, we adjust the values of A and R_{gb}. These steps are repeated until the calculated and measured effective thermal conductivities converge, resulting in fit values of A = 2 and R_{gb} = 0.1 m^2 K/GW. The fitted value of A = 2 indicates that the effective mean free path is twice that of the in-plane crystal size. This is reasonable because the cross-plane crystal size is much larger than the in-plane crystal size. R_{gb} is also close to the simulation values reported in refs 35 and 36. The good agreement between the measured effective thermal conductivities and the calculated gradient-model values using A = 2 and R_{gb} = 0.1 m^2 K/GW is depicted in Figure 5. On the basis of these two parameters, we calculate the gradient thermal conductivity along the membrane thickness direction.

It should be noted that the sensitivity of the Al−diamond interface conductance at the nucleation side when fitting the experimental data is very large at all thermal modulation frequencies (comparable to that of the diamond thermal conductivity for 1.2 MHz, about 5 times as large as that of the diamond thermal conductivity for 11.6 MHz). When fitting the

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TDTR data at each modulation frequency for both the Al–diamond interface conductance and the effective thermal conductivity of the diamond, the extracted value for the Al–diamond interface conductance consistently fit at 91 ± 2 MW/(m^2 K), with the small uncertainty attributed to the fact that the measurements at each modulation frequency were made at the same spot on the film. This is an important point in that even with the larger sensitivity to this parameter compared to that to the diamond thermal conductivity, the value for the Al–diamond interface conductance is independent of the thermal modulation frequency, thereby reinforcing the observed relation between the modulation frequency and the effective diamond thermal conductivity.

Figure 6 shows the local thermal conductivity of the 10 layers in the gradient thermal conductivity model assuming the fitted values for A and R_{gb}. The local thermal conductivity goes beyond 1000 W/(m K) when the distance from the nucleation interface only reaches 3 μm, which is higher than the values in the literature with similar thickness.[9,22−24,37] This confirms the excellent quality of this diamond membrane. The local thermal conductivity increases with distance from the nucleation interface. Near the nucleation interface, grain sizes are small and phonon scattering at the grain boundaries is the dominant mechanism that limits the phonon mean free path. The local thermal conductivity as a function of distance from the nucleation side of the film exhibits a nonlinear trend. Near the surface of the growth side, grain sizes are large and phonon–phonon (Umklapp) scattering is the dominant mechanism, which limits the phonon mean free path. However, as we move closer to the nucleation side of the membrane, the local thermal conductivity decreases as the increasingly smaller grains cause the phonons to scatter more often with grain boundaries than with themselves. All of the above analyses are based on a first-order approximation, and a more detailed analysis of the 3D anisotropic thermal conductivity will be discussed later.

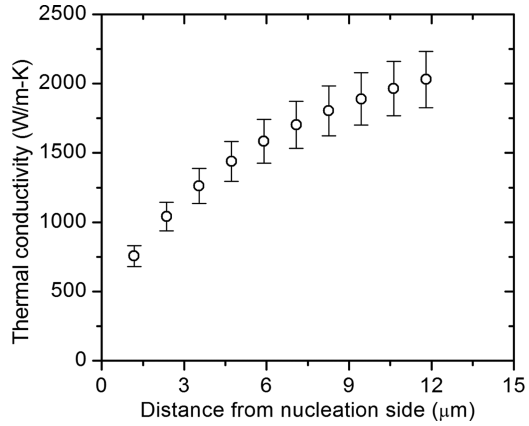


Figure 6. Gradient thermal conductivity of the diamond membrane.

3.2. Anisotropic Thermal Conductivity Measured by Different Spot Sizes. When the TDTR beam spot size is much larger than the penetration depth, heat transfers one dimensionally along the cross-plane direction and the sensitivity of the TDTR data for fitting the cross-plane thermal conductivity is much larger than that of in-plane thermal conductivity, as shown in Figure 7. When using the large spot sizes provided with the 5× objective (19.8 and 7.3 μm radii for the pump and probe, respectively), the sensitivity of κ_z is much larger than that of κ_r, allowing us to extract cross-plane thermal conductivity easily from the 5× objective measurement. At the same spot on the same sample with the same modulation frequency, we perform another TDTR measurement with a 20× objective (5.3 and 2.1 μm radii for the pump and probe, respectively). When using the smaller spot sizes produced by the 20× objective, heat transfers more radially and the sensitivity of the TDTR measurement to the in-plane thermal conductivity increases, whereas that to the cross-plane thermal conductivity decreases, as shown in Figure 7. Using these two different sets of spot sizes to take separate TDTR scans at the same location on the sample, we can iteratively determine both the in-plane and cross-plane thermal conductivities by finding values that fit both independent TDTR measurements. Our procedure is as follows: we first fit the cross-plane thermal conductivity using data collected with the 5× objective. We then insert this value for cross-plane thermal conductivity into the data fitting as a fixed parameter and then fit for the in-plane thermal conductivity using TDTR data taken at the same location with the 20× objective. This value for the in-plane thermal conductivity is input to and held fixed in the data fitting, the 5× TDTR data is then fitted for the cross-plane conductivity, and the iterative procedure is repeated until both the in-plane and cross-plane thermal conductivities fit well in both data sets. Therefore, with different spot sizes measured at the same spot with the same modulation frequency, we obtain both in-plane and cross-plane thermal conductivities.

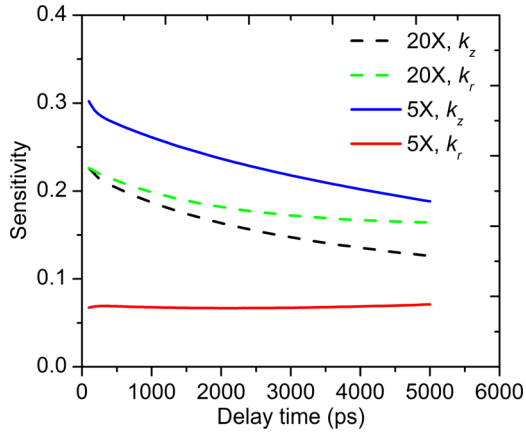


Figure 7. Sensitivity of in-plane and cross-plane thermal conductivities with 20× and 5× objectives. The pump and probe radii are 5.3 and 2.1 μm for the 20× objective and 19.8 and 7.3 μm for the 5× objective, respectively.

We performed TDTR measurements with 5× and 20× objectives for frequencies of 3.6 and 6.3 MHz. For smaller frequencies, the sensitivity of the TDTR data to the in-plane and cross-plane thermal conductivities with the 5× objective is comparable, and neither the cross-plane nor in-plane thermal conductivities can be extracted independently. For larger frequencies, the sensitivity of the data to the in-plane thermal conductivity when using the 20× objective is too small to obtain accurate in-plane thermal conductivity.[38] The measured cross-plane and in-plane thermal conductivities are shown in Table 1. The cross-plane thermal conductivities of 3.6 and 6.3 MHz are 1296 and 1182 W/(m K), respectively, whereas the corresponding in-plane thermal conductivities are 620 and 531 W/(m K), respectively. Whereas the values for the cross-plane thermal conductivity compare quite well between the two modulation frequencies used (about 10% difference), the in-

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Table 1. Cross-Plane and In-Plane Thermal Conductivities Measured with Different Spot Sizes for Different Modulation Frequencies

	κ_z (W/(m K))	κ_r (W/(m K))
3.6 MHz	1296	620
6.3 MHz	1182	531

plane thermal conductivities differ by as much as 17%. This is consistent with the explanation provided earlier in the manuscript, where the smaller penetration depth associated with the higher modulation frequency results in a greater fraction of the volume near the nucleation side being sampled, yielding lower effective cross-plane and in-plane thermal conductivities. It should again be mentioned that the fitted values for the Al−diamond interface conductance are very consistent with each other.

3.3. Three-Dimensional Anisotropic Thermal Conductivity. For 3D anisotropic thermal conduction in the CVD diamond membrane, we used the thermal model in ref 9. In-plane grain size (d_r) is modeled as $d_r(z) = d_0 + \alpha \times z$, where d_0 is the crystal size at the nucleation interface, α is a constant, and z is the distance from the nucleation interface. Cross-plane crystal size (d_z) is given by

$$\frac{d_z(z)}{L} = \left(\frac{z}{L} + \frac{\beta}{\alpha}\right) \times \left[\frac{1}{\log(g)} \log\left(\frac{\alpha + \beta}{\alpha z/L + \beta}\right) + 1\right] - \frac{\beta}{\alpha} \quad (4)$$

where $\beta = d_0/L$ and g is the inverse survival rate, which in this work we fix as 2.[9] Taking into consideration several types of phonon scattering mechanisms, including phonon−phonon, as well as in-plane and cross-plane grain boundary scattering, the effective mean free path in the cross-plane and in-plane directions can be obtained by[9,39,40]

$$\lambda_{z,r}^{-1}(z) = \frac{1}{\lambda_{bulk}} + \frac{1}{B \times d_{z,r}(z)} + \frac{1}{C \times d_{r,z}(z)} \quad (5)$$

where B and C are constants that are related to the probability of phonon transmission t and the specularity parameter p, respectively. Assuming a gray (frequency-independent) approximation with regard to these parameters, $B = 0.75t/(1 - t)$ and $C = (1 + p)/(1 - p)$ represent the cases where heat transport is limited by scattering that is perpendicular to interfaces (t effect) and parallel to interfaces (p effect). After obtaining the in-plane and cross-plane effective phonon mean free paths, the in-plane and cross-plane thermal conductivities can be obtained by inserting them into eqs 2 and 3, assuming $R_{gb} = 0.1$ m^2 K/GW. Finally, we can fit t and p using TDTR measurements collected at 3.6 and 6.3 MHz, using both the 5× and 20× objectives. The diamond membrane is divided into 10 layers, as shown in Figure 4. For each layer, we can obtain the in-plane and cross-plane thermal conductivities through an iterative method similar to that described previously: first we make a guess for the values of t and p, then we generate a theoretical TDTR−V_{in}/V_{out} curve, which we can compare to the experimental data, and then revise our guesses for t and p to repeat the process. Finally, we obtain t as 0.56 and p as 0.33, which are reasonable for relatively high-quality diamond grain boundaries. Figure 8 shows the good agreement between TDTR experimental data and theoretical curves. The 3D anisotropic thermal conductivities (change in the in-plane and cross-plane thermal conductivities with distance from the nucleation interface) are shown in Figure 9. Both the in-plane and cross-plane thermal conductivities increase significantly with increasing distance from the nucleation interface because thermal transport transitions from being grain boundary scattering limited to phonon−phonon scattering limited as the grains get larger. If predicting the thermal conductivity with this thermal model, both the local in-plane and cross-plane thermal conductivities of this diamond membrane would go beyond 2000 W/(m K) with an anisotropy ratio of 20% when the distance from the nucleation interface reaches 26 μm. This high-quality diamond facilitates heat dissipation significantly.

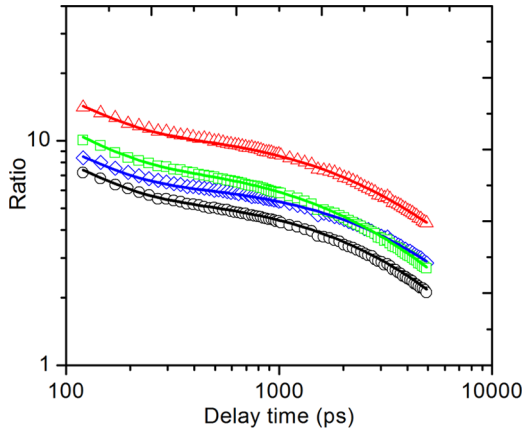


Figure 8. Experimental measurements are fitted with a theoretical model. The scatters are experimental data, whereas the lines are fitting curves. Blue is data with 5× objective and 3.6 MHz. Red is data with 20× objective and 3.6 MHz. Black is data with 5× objective and 6.3 MHz. Green is data with 20× objective and 6.3 MHz.

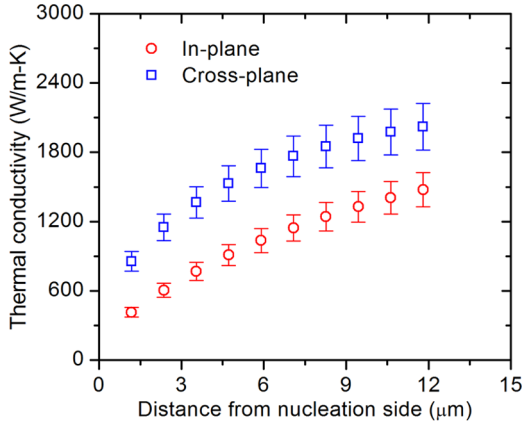


Figure 9. In-plane and cross-plane thermal conductivities along the cross-plane direction.

We attribute the frequency and spot-size-dependent thermal conductivity to the anisotropic structure of CVD diamond. We noticed that, in TDTR measurements, ballistic thermal transport results in reduced thermal conductivity when laser spot size or penetration depth is smaller than or comparable to phonon mean free path because phonons with mean free path larger than spot size or penetration depth do not contribute to thermal conduction.[41−43] In bulk diamond, at room temperature phonons with mean free path from 550 to 3400 nm contribute to 80% of heat conduction.[34] In CVD diamond, especially the nucleation side of the membrane, the small

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crystal size limits the phonon mean free path, making ballistic thermal transport more difficult to be observed. In our measurements, the root-mean spot sizes for 5× and 20× are 14.9 and 4.0 μm, respectively. In ref 36 reduction in thermal conductivity can be observed only if the root-mean spot size is smaller than 2 μm for bulk diamond. This limit would be much smaller for CVD polycrystalline diamond membrane. Thus, ballistic thermal transport should not show up in measurements with these spot sizes. In terms of modulation frequency, we can estimate the penetration depth with the formula $\sqrt{\alpha_t/\pi f}$, where α_t is thermal diffusivity of diamond and f is modulation frequency in TDTR measurements. The smallest penetration depth in this paper is around 3.5 μm, which is much larger than the possible phonon mean free path in the CVD diamond. Therefore, ballistic thermal transport does not affect our TDTR measurements to probe the inhomogeneous thermal conduction in CVD diamond. Additionally, all of the error bars reported in this work are estimated to be ±10% for general TDTR measurements.

4. CONCLUSIONS

Owing to the inhomogeneous structure arising during the CVD growth process, the thermal conductivities of these diamond films are highly anisotropic in layers near the nucleation and growth interface. This anisotropy can be probed using thermoreflectance methods by modulating the measurement frequency and spot size, combined with appropriate models to address the spatial gradient in thermal properties. In this work, TDTR was used to measure thermal properties of an 11.8 μm CVD diamond membrane from its nucleation side. By changing the modulation frequencies from 1.2 to 11.6 MHz to tune the heat penetration depth and subsequently the part of diamond sampled by TDTR, we obtained a 2D gradient thermal conductivity along the CVD diamond growth direction. The local thermal conductivity goes beyond 1000 W/(m K) when the distance from the nucleation interface only reaches 3 μm. In addition, by measuring the same spot on the sample with different spot sizes, we were able to independently determine the in-plane and cross-plane thermal conductivities. Through a combination of using multiple spot sizes and multiple modulation frequencies (3.6 and 6.3 MHz), the full 3D anisotropic thermal conductivity of a CVD diamond membrane was experimentally obtained. Predicted by our model, both the local in-plane and cross-plane thermal conductivities of this diamond membrane would go beyond 2000 W/(m K) with an anisotropy ratio of 20% when the distance from the nucleation interface reaches 26 μm. Our work provides an improved understanding of thermal conductivity inhomogeneity in high-quality CVD polycrystalline diamond that is important for applications in thermal management of high-power electronics.

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Notes

The authors declare no competing financial interest.

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